



- (a) (i) Put a tick (✓) in the correct column to show whether the following statements apply to Mendeleev's table only, today's table only or to both tables. [2]

|                                             | Mendeleev<br>only | Today<br>only | Both<br>tables |
|---------------------------------------------|-------------------|---------------|----------------|
| the table is organised into groups          |                   |               |                |
| copper and potassium are in the same group  |                   |               |                |
| there are gaps in the table                 |                   |               |                |
| fluorine and chlorine are in the same group |                   |               |                |

- (ii) Tick (✓) the **two** statements that **best** describe why germanium was confirmed to be the element ekasilicon predicted by Mendeleev. [1]

germanium has exactly the same atomic mass as that predicted for ekasilicon

☐

germanium has a different colour to that predicted for ekasilicon

☐

germanium has a similar density to that predicted for ekasilicon

☐

germanium oxide has the same ratio of atoms as that predicted for ekasilicon oxide

☐

germanium oxide and germanium chloride have the same ratio of atoms

☐


- (b) (i) Calculate the percentage of oxygen present in the formula of germanium oxide,  $\text{GeO}_2$ . [2]

$$A_r(\text{Ge}) = 73$$

$$A_r(\text{O}) = 16$$

Percentage = ..... %

- (ii) When germanium oxide reacts with hydrochloric acid it produces germanium chloride and water.

Balance the following equation for the reaction that takes place. [1]

